



Quantum signals transmitted over commercial fibre network

Researchers at the University of Pennsylvania (Philadelphia, Pennsylvania) have transmitted quantum signals over Verizon's commercial fibre network using standard



A node of the quantum network, roughly one kilometre's worth of Verizon fibre-optic cable away from the quantum signal's source (Credit: <https://phys.org>)

Internet Protocol (IP). Central to the breakthrough is the Q-Chip, which couples classical 'headers' with quantum data, enabling routing, switching, and noise correction without disturbing quantum states. Tested across one kilometre with 97% fidelity, the system demonstrates scalable quantum networking on existing infrastructure, addressing real-world noise challenges though amplification beyond metro ranges remains unresolved.

SPAES enables lighter, more responsive XR devices

VoxelSensors (City of Brussels, Belgium) is addressing key XR challenges, including power drain, latency, and unreliable sensors, with its single photon active event sensor (SPAES) technology. Using up to 10x less power than standard depth sensors, SPAES ensures low-latency, stable performance across environments, enabling lighter, more



VoxelSensors to advance next-generation depth sensing technology with 10x power savings for XR applications (Credit: <https://voxelsensors.com/news>)

responsive XR devices. Integrated with Qualcomm's Snapdragon AR2 Gen 1 Platform, it powers applications like eye tracking and personal AI agents, paving the way for practical, intelligent, and energy-efficient AR/VR systems.

Salt could supercharge next-gen solar cells

Researchers at University College London (England) have discovered that a common salt, guanidinium thiocyanate, can enhance both efficiency and durability in perovskite

solar cells. By slowing crystal growth during fabrication, the additive produces smoother, defect-free layers, enabling a record 22.3% efficiency in mixed tin-lead perovskites—close to silicon's lab best. This simple approach promises longer-lasting, high-performance perovskite tandems, advancing flexible, low-cost, next-generation solar technology.

A new elastomer converts body heat into electricity

Scientists at Peking University (Beijing, China) have developed a rubber-like thermoelectric elastomer that converts body heat into electricity, paving the way for self-powered wearables and medical implants. Reported in *Nature*, it is the first highly elastic n-type material combining skin-like stretchability with efficient energy conversion. Retaining performance when stretched up to 150% and enduring extreme strain, the innovation could eliminate bulky batteries, powering health patches, fitness trackers, smart textiles, and implantable devices sustainably.

A silicon photonic chip enhances AI efficiency

AI's growing power needs are pushing chip innovation. Researchers at the University of Florida (University of Florida) have built a silicon photonic chip that uses laser light instead of electricity to perform convolution, the core of AI pattern recognition, at up to 100 times higher efficiency. By integrating ultrathin Fresnel lenses on silicon, the chip executes optical convolutions, achieving 98% accuracy in handwritten digit recognition at near-zero energy. Supporting wavelength multiplexing, it promises greener, scalable AI hardware.



A newly developed silicon photonic chip turns light-encoded data into instant convolution results (Credit: H. Yang, University of Florida)

3D-printed metamaterial enhances tactile sensor performance

Researchers at Seoul National University of Science and Technology (Seoul, South Korea) have developed a 3D-printed auxetic metamaterial that significantly enhances tactile sensor performance for wearables, robotics, and healthcare applications. Built with a cubic lattice of spherical voids, the design boosts sensitivity, stability, and customisation while minimising interference. Tested in both capacitive and piezoresistive modes, the platform demonstrated high-fidelity gait and pressure mapping capabilities. Scalable and adaptable, it opens the path to next-generation, mass-customised sensing in prosthetics, rehabilitation, and smart devices.